



Title :

Week 3 : SOC Operations, Threat Intelligence, and Incident Escalation Report

Introduction :

This report documents the theoretical concepts and practical activities performed during Week 3 SOC training. The focus areas included advanced log analysis, threat intelligence integration, incident escalation workflows, alert triage, evidence preservation, and a complete SOC workflow simulation. Industry-standard frameworks and platforms such as MITRE ATT&CK, AlienVault OTX, VirusTotal, Elastic Security, and Metasploit were studied to understand modern Security Operations Center practices.

Objective :

The main objectives of this task were :

- To understand how security analysts correlate logs from multiple sources to identify suspicious activities and reduce false positives.
- To identify unusual behavior that may indicate malicious activity.
- To enrich security alerts using external threat intelligence sources.
- To understand how incidents are escalated between SOC tiers.
- To communicate incident details clearly to stakeholders.
- To prioritize and investigate security alerts based on risk and impact.
- To validate indicators of compromise using external intelligence sources.
- To collect and preserve digital evidence while maintaining integrity.
- To maintain accountability for collected evidence.
- To understand vulnerability assessment and exploitation concepts.
- To simulate a complete alert-to-response workflow.
- To visualize the complete SOC operational process.



Advanced Log Analysis :

Log Correlation Example :

Timestamp	Event ID	Source IP	Destination IP	Notes
2025-08-18 12:00	4625	192.168.1.100	8.8.8.8	Suspicious DNS Request

Screenshots :

Timestamp	Event ID	Source IP	Destination IP	Notes
2025-08-18 12:00	4625	192.168.2.10	8.8.8.8	Suspicious DNS

Fig.1 : Elastic Security and Log Correlation Table

Security analysts use Elastic Security to correlate events from multiple sources and identify suspicious activities. This table demonstrates log correlation by linking failed login attempts with network activity.



Anomaly Detection :

User	Login Time
Administrator	02:00 AM
Employee	09:00 AM

Screenshot :

User	Login Time
Admin	02:00 AM
Employee	09:00 AM

Fig. 2 : Anomaly Detection

This example illustrates anomaly detection by identifying unusual login behavior. Security monitoring systems use anomaly detection techniques to discover activities that differ from normal user behavior.



Threat Intelligence Integration :

Threat Intelligence Sources Studied :

- AlienVault OTX
- MITRE ATT&CK
- VirusTotal

Screenshots :

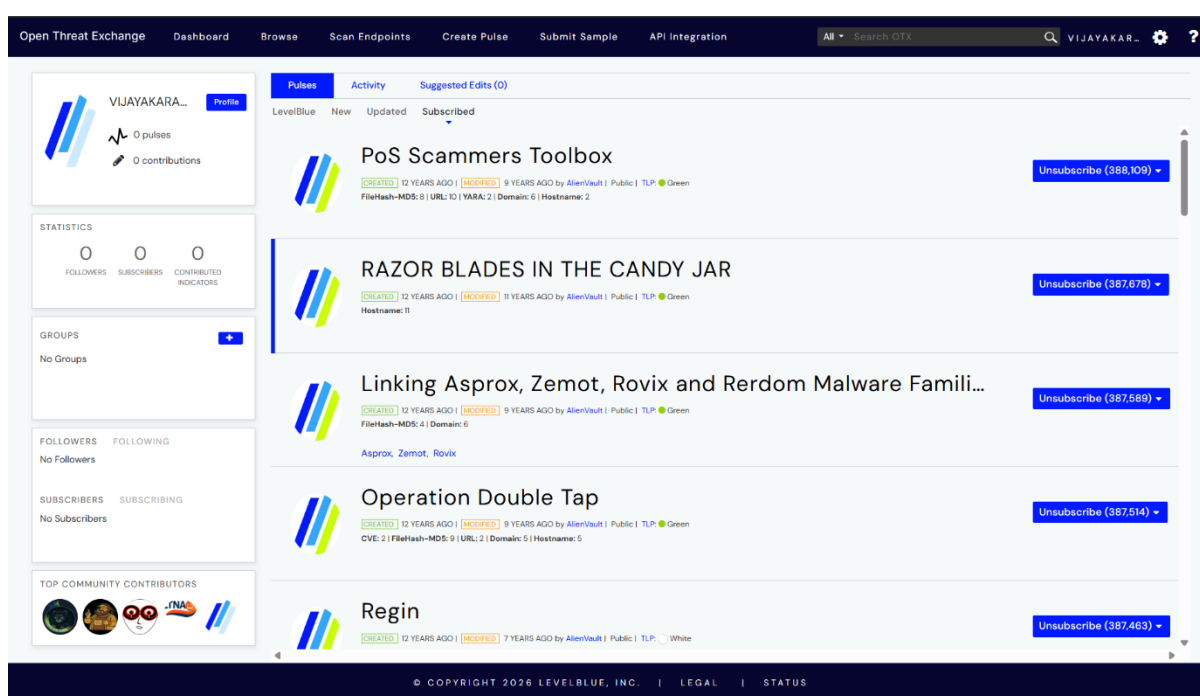


Fig.3 : AlienVault OTX

AlienVault Open Threat Exchange (OTX) is a threat intelligence platform that provides indicators of compromise (IOCs), malicious IP addresses, domains, and threat reports. Analysts use OTX to enrich alerts and validate suspicious activity.



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MITRE ATT&CK® is a globally-accessible knowledge base of adversary tactics and techniques based on real-world observations. The ATT&CK knowledge base is used as a foundation for the development of specific threat models and methodologies in the private sector, in government, and in the cybersecurity product and service community.

With the creation of ATT&CK, MITRE is fulfilling its mission to solve problems for a safer world – by bringing communities together to develop more effective cybersecurity. ATT&CK is open and available to any person or organization for use at no charge.

ATT&CK Matrix for Enterprise

layout: side ▾ show sub-techniques hide sub-techniques

Resource Development	Initial Access	Execution	Persistence	Privilege Escalation	Stealth	Defense Impairment	Credential Access	Discovery	Lateral Movement	Collection	Communication
9 techniques	11 techniques	20 techniques	22 techniques	13 techniques	30 techniques	18 techniques	17 techniques	34 techniques	9 techniques	17 techniques	18 techniques
Acquire Access	Content Injection	BITS Jobs	Account Manipulation (7)	Abuse Elevation Control Mechanism (5)	Access Token Manipulation (5)	Disable or Modify System Firewall (3)	Adversary-in-the-Middle (4)	Account Discovery (4)	Exploitation of Remote Services	Adversary-in-the-Middle (4)	Application Layer Protocol Manipulation (1)
Acquire Infrastructure (6)	Drive-by Compromise	Cloud Administration Command	BITS Jobs	Access Token Manipulation (5)	Build Image on Host	Disable or Modify Tools (5)	Brute Force (4)	Application Window Discovery	Internal Spearphishing	Archive Collected Data (3)	Communications Through Removable Media (1)
Compromise Accounts (9)	Exploit Public-Facing Application	Command and Scripting Interpreter (13)	Boot or Logon Autostart Execution (14)	Account Manipulation (7)	Debugger Evasion	Domain or Tenant Policy Modification (2)	Credentials from Password Stores (5)	Browser Information Discovery	Lateral Tool Transfer	Audio Capture	Content Injection
Compromise Infrastructure (6)	External Remote Services	Container Administration Command	Boot or Logon Initialization Scripts (5)	Boot or Logon Autostart Execution (14)	Delay Execution	Downgrade Attack	Exploitation for Credential Access	Cloud Infrastructure Discovery	Remote Service Session Hijacking (2)	Automated Collection	Data Encoding
Develop Capabilities (4)	Hardware Additions	Deploy Container	Cloud Application Integration	Boot or Logon	Deobfuscate/Decode Files or Information	Exploitation for Defense	Forced Authentication	Cloud Service Dashboard	Remote	Browser Session Hijacking	Data
Establish Accounts (9)					Direct Volume Access			Cloud Service Discovery			

Fig.4 : MITRE ATT&CK

The MITRE ATT&CK framework is a globally recognized knowledge base of adversary tactics and techniques. It helps SOC analysts classify attacks, perform threat hunting, and map incidents to known attacker behaviors.



Alert Enrichment Example :

Alert ID	IP Address	Reputation	Notes
003	192.168.1.100	Malicious	Linked to C2 Infrastructure

Screenshot :

Alert ID	IP	Reputation	Notes
003	192.168.1.100	Malicious	Linked to C2

Fig.5 : Threat Enrichment

This table demonstrates alert enrichment using threat intelligence sources. Enrichment provides additional context such as reputation information, helping analysts prioritize investigations and reduce false positives.



Incident Escalation Workflow :

SOC Escalation Structure :

- Tier 1: Alert Monitoring and Triage
- Tier 2: Incident Investigation
- Tier 3: Advanced Analysis and Threat Hunting

Screenshot :

Incident Escalation Summary :

Incident:

- Unauthorized Access

Source IP:

- 192.168.1.200

Priority:

- High

|

Action:

- Escalated to Tier 2

Fig.6 : Incident Escalation

This screenshot shows a sample incident escalation process. Escalation ensures that high-priority security incidents are transferred from Tier 1 analysts to more experienced analysts for deeper investigation.



Situation Report (SITREP) :

Incident Summary :

Title: Unauthorized Access on Server-Y

Source IP: 192.168.1.200

MITRE Technique: T1078 (Valid Accounts)

Actions Taken:

- Server isolated
- Investigation initiated
- Escalated to Tier 2

Screenshot :

Title:

Unauthorized Access on Server-Y

Summary:

- Detected suspicious login activity.

Affected Host:

- Server-Y

Source IP:

- 192.168.1.200

Action Taken:

- ✓ Server isolated.
- ✓ Incident escalated.

Fig.7 : Situation Report

The Situation Report (SITREP) summarizes incident details, affected systems, indicators of compromise, and actions taken. It provides stakeholders with a clear overview of the current security situation.



Alert Triage Practice :

Alert Example :

Alert ID	Description	Source IP	Priority	Status
004	Suspicious PowerShell Execution	192.168.1.101	High	Open

Screenshot :

Alert ID	Description	Source IP	Priority	Status
004	PowerShell Execution	192.168.1.101	High	Open

Fig.8 : Alert Triage

This screenshot illustrates the alert triage process. Alerts are reviewed, prioritized, and categorized based on severity, impact, and likelihood to ensure efficient incident response.



IOC Validation Using Threat Intelligence :

VirusTotal Analysis :

Screenshot :

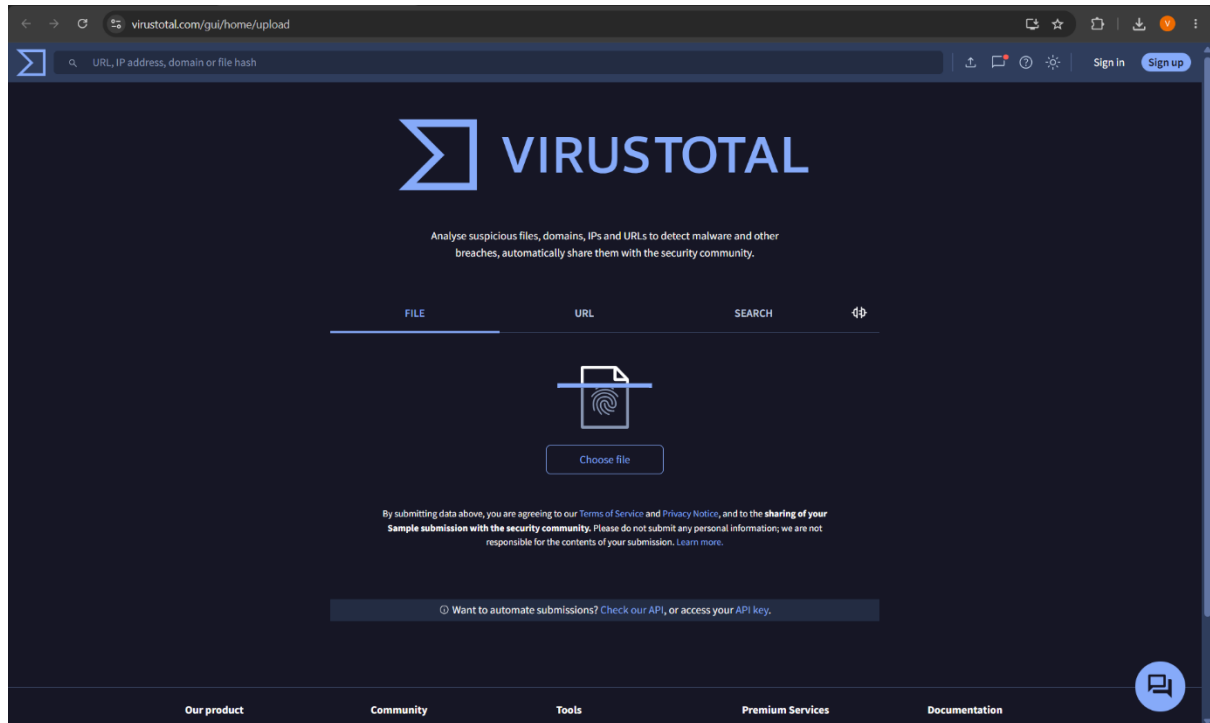


Fig.9 : VirusTotal Lookup

VirusTotal is used to analyze suspicious files, URLs, domains, and IP addresses by comparing them against multiple security engines. It helps validate indicators of compromise during investigations.



Evidence Preservation :

Evidence Collection Example :

Item	Description	Collected By	Date	Hash Value
Memory Dump	Server-Y Memory	SOC Analyst	05-06-2026	SHA256 Example

Screenshot :

Item	Description	Collected By	Date	Hash
Memory Dump	Server-Y	SOC Analyst	05-06-2026	SHA256

Fig.10 : Evidence Preservation

This screenshot demonstrates evidence collection and preservation procedures. Proper evidence handling ensures forensic data remains accurate and admissible throughout an investigation.



Chain of Custody :

Evidence	Handler	Time
Memory Dump	SOC Analyst	10:00 AM

Screenshot :

Evidence	Handler	Time
Memory Dump	SOC Analyst	10:00 AM

Fig.11 : Chain of Custody

Chain of custody records document who collected, accessed, and handled evidence.

Maintaining a proper chain of custody preserves evidence integrity and accountability.



Metasploit Security Testing :

Metasploit Console :

```
kali@kali:~$ msfconsole
Session Actions Edit View Help
msf> search samba
Metasploit tip: Writing a custom module? After editing your module, why not try
the reload command

Metasploit v6.4.126-dev
+ -- 2,638 exploits - 1,331 auxiliary - 2,153 payloads
+ -- 432 post - 49 encoders - 15 nops - 12 evasion

Metasploit Documentation: https://docs.metasploit.com/
The Metasploit Framework is a Rapid Open Source Project

msf> search samba
Matching Modules
#  Name                                     Disclosure Date  Rank  Check  Description
-  -
0  exploit/unix/webapp/citrix_access_gateway_exec 2010-12-21      excellent Yes  Citrix Access Gateway Command Execution
1  exploit/windows/licenses/calliclnt_getconfig 2005-03-02      average No   Computer Associates License Client GETCONFIG Overflow
2  \target: Automatic
3  \target: Windows 2000 English
4  \target: Windows XP English SP0-1
5  \target: Windows XP English SP2
6  \target: Windows 2003 English SP0
7  exploit/unix/misc/distcc_exec 2002-02-01      excellent Yes  DistCC Daemon Command Execution
8  exploit/windows/smb/group_policy_startup 2015-01-26      manual No   Group Policy Script Execution From Shared Resource
9  \target: Windows x86
10 \target: Windows x64
11 post/linux/gather/enum_configs - - - - - Linux Gather Configurations
12 auxiliary/scanner/rpcnsc/modbus_list - - - - - List Modbus Modules
13 exploit/windows/fileformat/ms14_060_sandworm 2014-10-14      excellent No   MS14-060 Microsoft Windows OLE Package Manager Code Execution
14 exploit/unix/http/quest_kace_systems_management_vce 2010-09-28      excellent Yes  Quest KACE Systems Management Command Injection
15 exploit/multi/smb/usermap_script 2007-05-14      excellent No   Samba "username map script" Command Execution
16 exploit/multi/smb/nttrans 2003-04-07      average No   Samba 2.2.2 - 2.2.5 nttrans Buffer Overflow
17 exploit/linux/smb/setinformationpolicy_auditinfo_heap_overflow 2012-04-10      normal Yes  Samba SetInformationPolicy AuditEventsInfo Heap Overflow
18 \target: 213.5.11-dfsg-lubuntu2 on Ubuntu Server 11.10
19 \target: 213.5.8-dfsg-lubuntu2 on Ubuntu Server 11.10
20 \target: 213.5.8-dfsg-lubuntu2 on Ubuntu Server 11.04
21 \target: 213.5.4-dfsg-lubuntu8 on Ubuntu Server 10.10
22 \target: 213.5.4-dfsg-squeeze8 on Debian Squeeze
23 \target: 315.10-107-015 on CentOS 5
24 auxiliary/admin/smb/smb_symlink_traversal - - - - - Samba Symlink Directory Traversal
25 auxiliary/scanner/smb/smb_uninit_cred - - - - - Samba smb2_ServerPasswordSet Uninitialized Credential State
26 exploit/linux/smb/smb_chain_reply 2010-06-16      good No   Samba chain_reply Memory Corruption (Linux x86)
27 \target: Linux (Debian 3.2.5-4lenny6)
28 \target: Subverting Target
29 exploit/linux/smb/is_known_pipename 2017-03-24      excellent Yes  Samba is_known_pipename() Arbitrary Module Load
30 \target: Automatic (Interact)
31 \target: Automatic (Command)
32 \target: Linux x86
33 \target: Linux x86_64
```

Fig.12 : Metasploit Console

This screenshot displays the Metasploit Framework console, which is widely used for penetration testing, vulnerability assessment, and security research in controlled environments.

Module Search :

Command Used :

- *search samba*

Exploit Module :

Command Used :

- *use exploit/multi/samba/usermap_script*



```
Session Actions Edit View Help
26 exploit/linux/samba/chain_reply 2010-06-16 good No Samba chain_reply Memory Corruption (Linux x86)
27 \ target: Linux (Debian 3.2.5-4lemmy6)
28 \ target: Subverting Target
29 exploit/linux/samba/is_known_pipename 2017-03-24 excellent Yes Samba is_known_pipename() Arbitrary Module Load
30 \ target: Automatic (Interact)
31 \ target: Automatic (Command)
32 \ target: Linux x86
33 \ target: Linux x86_64
34 \ target: Linux ARM (LE)
35 \ target: Linux ARMv4
36 \ target: Linux MIPS
37 \ target: Linux MIPSLE
38 \ target: Linux MIPS64
39 \ target: Linux MIPS64LE
40 \ target: Linux PPC
41 \ target: Linux PPC64
42 \ target: Linux PPC64 (LE)
43 \ target: Linux SPARC
44 \ target: Linux SPARC64
45 \ target: Linux s390x
46 auxiliary/dos/smb/lsa_addprivs_heap normal No Samba lsa_io_privilege_set Heap Overflow
47 auxiliary/dos/smb/lsa_transnames_heap normal No Samba lsa_io_trans_names Heap Overflow
48 exploit/linux/samba/lsa_transnames_heap 2007-05-14 good Yes Samba lsa_io_trans_names Heap Overflow
49 \ target: Linux Syscall
50 \ target: Linux Heap Brute Force (Debian/Ubuntu)
51 \ target: Linux Heap Brute Force (Gentoo)
52 \ target: Linux Heap Brute Force (Mandriva)
53 \ target: Linux Heap Brute Force (RHEL/CentOS)
54 \ target: Linux Heap Brute Force (SUSE)
55 \ target: Linux Heap Brute Force (Slackware)
56 \ target: Linux Heap Brute Force (OpenWRT MIPS)
57 \ target: NetBSD
58 exploit/os/smb/lsa_transnames_heap 2007-05-14 average No Samba lsa_io_trans_names Heap Overflow
59 \ target: Automatic
60 \ target: Mac OS X 10.4.x x86 Samba 3.0.10
61 \ target: Mac OS X 10.4.x PPC Samba 3.0.10
62 \ target: NetBSD
63 exploit/solaris/smb/lsa_transnames_heap 2007-05-14 average No Samba lsa_io_trans_names Heap Overflow
64 \ target: Solaris 8/9/10 x86 Samba 3.0.21-3.0.24
65 \ target: Solaris 8/9/10 SPARC Samba 3.0.21-3.0.24
66 \ target: NetBSD
67 auxiliary/dos/smb/read_nttrans_ea_list 2003-04-07 normal No Samba read_nttrans_ea_list Integer Overflow
68 exploit/freebsd/smb/trans2open 2003-04-07 great No Samba trans2open Overflow (4BSD x86)
69 exploit/linux/smb/trans2open 2003-04-07 great No Samba trans2open Overflow (Linux x86)
70 exploit/os/smb/trans2open 2003-04-07 great No Samba trans2open Overflow (Mac OS X PPC)
71 exploit/solaris/smb/trans2open 2003-04-07 great No Samba trans2open Overflow (Solaris SPARC)
72 \ target: Samba 2.2.x - Solaris 9 (sun4u) - Bruteforce
73 \ target: Samba 2.2.x - Solaris 7/8 (sun4u) - Bruteforce
74 exploit/windows/http/smbsearch_results 2003-06-21 normal Yes Samba 6 Search Results Buffer Overflow
75 \ target: Automatic
76 \ target: Windows 2000
77 \ target: Windows XP

Interact with a module by name or index. For example info 77, use 77 or use exploit/windows/http/smbsearch_results
After interacting with a module you can manually set a TARGET with set TARGET windows XP

msf > use exploit/multi/samba/usermap_script
[+] No payload configured, defaulting to cmd/unix/reverse_netcat
msf exploit(multi/samba/usermap_script) > run
[*] MetasploitOptionValidateError One or more options failed to validate: RHOSTS.
msf exploit(multi/samba/usermap_script) > 
```

Fig.13 : Metasploit Samba

This screenshot shows the search results for Samba-related exploit modules within Metasploit. Security professionals use these modules to understand vulnerabilities and test defenses. This screenshot demonstrates the configuration of the Samba Usermap Script exploit module. The exercise helps analysts understand how attackers may exploit vulnerable services.



SOC Incident Response Simulation :

Incident Summary :

Attack Type:

- Samba Usermap Script Exploit

Detection:

- SOC Alert Generated

Containment:

- Attacker IP Blocked

Escalation:

- Tier 2 Analyst

Status:

- Resolved

Screenshot :

Attack:

- Samba Usermap Script Exploit

Detection:

- SOC Alert Generated

Containment:

- Attacker IP Blocked

Escalation:

- Tier 2 Analyst

Status:

- Resolved

Fig.14 : Incident Response Summary



SOC Workflow Diagram :

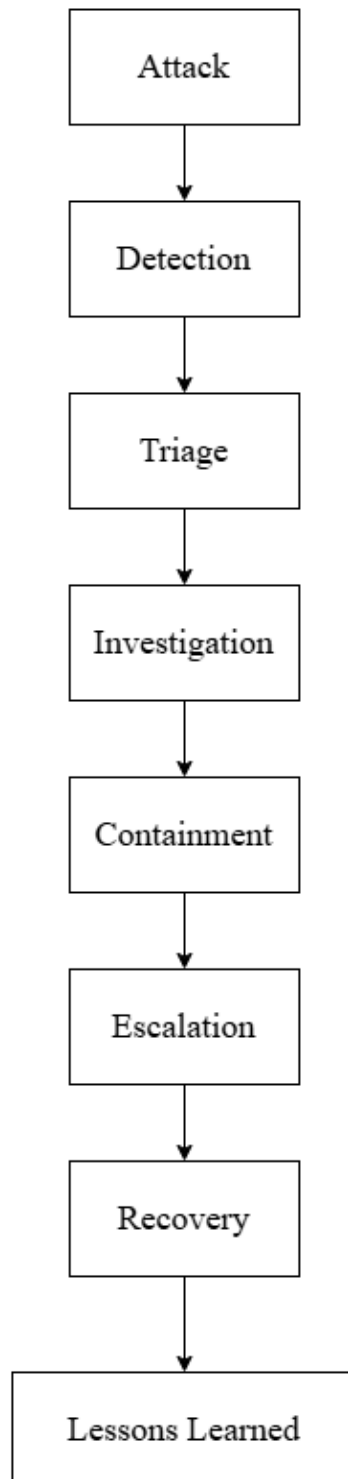


Fig.12 : SOC Workflow



Key Learnings :

- Learned how advanced log analysis and correlation techniques help identify complex attack patterns.
- Understood the role of threat intelligence platforms in enriching alerts and validating indicators of compromise.
- Gained practical knowledge of incident escalation, evidence preservation, alert triage, and SOC workflows.

Tools Studied :

- Elastic Security
- MITRE ATT&CK
- AlienVault OTX
- VirusTotal
- Metasploit Framework

Conclusion :

This exercise provided a comprehensive understanding of SOC operations, advanced log analysis, threat intelligence integration, incident escalation procedures, evidence handling, and incident response workflows. The activities demonstrated how modern SOC teams detect, investigate, contain, and document security incidents while leveraging threat intelligence and industry-standard methodologies.

References :

- <https://attack.mitre.org/>
- <https://otx.alienvault.com/>
- <https://www.virustotal.com/>
- <https://www.elastic.co/security>
- <https://docs.metasploit.com/>
- <https://csrc.nist.gov/publications/detail/sp/800-61/rev-2/final>
- <https://www.sans.org/>